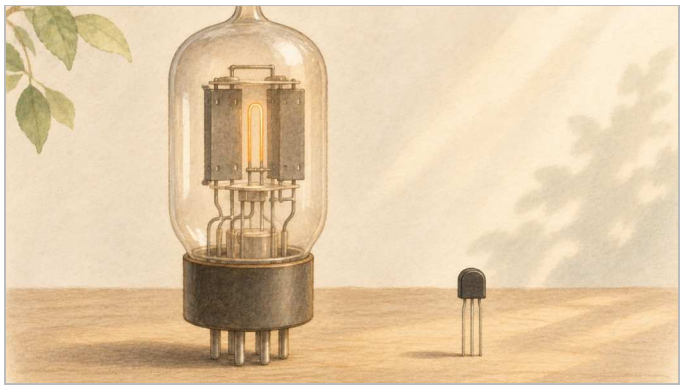
Transistors

01 What came before transistors?



A hot glass tube, and the tiny part that replaced it.

Look closer. Compare the tall glass tube with the tiny black bead beside it.

A little explanation. Glass vacuum tubes. A tube was about as tall as a hand. It glowed, got hot and burned out like an old light bulb. One early computer needed about 18,000 tubes and filled a room. In 1947, three scientists built the first transistor from gold foil pressed onto a small crystal.

Try this. Hold up your hand: an old tube was about that tall. Now look at your smallest fingernail. A transistor you can hold is smaller still. Find one thing in the room the size of each.

02 How can a tiny push control a big flow?



A small turn lets a big flow through.

Look closer. Look at how little the tap handle has turned, and how big the stream is.

A little explanation. A transistor has three legs. Electricity flows in one leg and out another, and the middle leg works like a tap handle. A tiny signal there opens the path for a much bigger flow. The name joins two words: transfer and resistor.

Try this. At a sink, turn the tap handle just a little. Watch how a small turn lets a big stream through. Then turn it back and watch the stream stop.

03 How can on-off switches count?



Light side on, dark side off: a row of switches.

Look closer. Count the light cards and the dark cards in the row.

A little explanation. A transistor switch is on or off, and a computer treats on as one and off as zero. Each extra switch doubles the patterns. Eight switches can make 256 patterns, enough for every letter on a keyboard.

Try this. Lay two paper scraps in a row: light side up is on, dark side up is off. How many different rows can you make? Add a third scrap and count again. Did the number double?

04 How do billions fit on one chip?



Too small to see: billions of tiny switches.

Look closer. Look through the lens at the tiny pattern on the chip.

A little explanation. They are printed, not built one by one. Light shines through a stencil onto a silicon wafer, layer after layer. In many tiny transistors, the middle part never touches the path. It sits behind a thin wall and steers the flow with an unseen electric push.

Try this. Cut a simple shape out of paper to make a stencil. Lay it on another sheet and rub a crayon across it. Print it three times. Chip makers print with light in a similar way.